Déjà

Cognee-Powered Personal Memory Recovery Engine

Build your external brain. Query your past. Recover the forgotten.

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CMR College of Engineering & Technology, Hyderabad

SDG 3: Good Health & Well-Being | SDG 9: Industry, Innovation & Infrastructure

PROBLEM STATEMENT

The crisis of cognitive overload in modern life

"Humans forget ~70% of new information within 24 hours — and there is no search engine for your own mind."

THE PROBLEM

- No personal semantic memory layer for individuals
- Journals and notes lack intelligent cross-referencing
- Forgotten habits, contradictions & patterns stay invisible
- Existing AI tools have no personalized longitudinal context

OUR SOLUTION — DÉJÀ

- Semantic knowledge graph built from your own journal entries
- Natural-language Q&A over your entire personal history
- Automated contradiction & gap detection (Blindspots)
- Dual local + cloud architecture — private & scalable

1. CUSTOMER SEGMENTS

Who are we creating value for?

Students & Job Seekers

- Track study habits, mood-performance links
- Recall preparation timelines for exams/interviews
- Identify study pattern gaps before deadlines

Knowledge Workers & Researchers

- Journal-based project memory for long-term work
- Semantic retrieval across months of notes & ideas
- Contradiction detection in research hypotheses

Mental Wellness & Self-Improvement

- Habit tracking with intelligent pattern analysis
- Reflect on behavioural trends over months & years
- Privacy-first local or cloud graph storage

Primary Market: Tech-savvy individuals aged 18-35 who journal digitally and value AI-augmented self-awareness.

2. UNIQUE VALUE PROPOSITION (UVP)

Why Déjà stands out as the ultimate cognitive companion

Dynamic Hybrid Indexing

- Bridges time-series timelines (Supabase) with semantic graph networks (Cognee).
- Maintains absolute chronological accuracy while enabling deep concept relationships.
- Constructs a living external brain layer of experiences.

Paradox & Gap Detection

- Our proprietary Blindspots Engine compares memory graph density with timeline logs.
- Automatically detects abandoned habits, routine shifts, and semantic contradictions.
- Surfaces insights you forgot to remember.

Memory Sanctuary

- Provides target node & edge pruning in the vector space via cognee.forget().
- True GDPR-aligned 'right to be forgotten' at a conceptual level.
- Users choose what the AI retains, remembers, or permanently dissolves.

10× Accelerated Recall

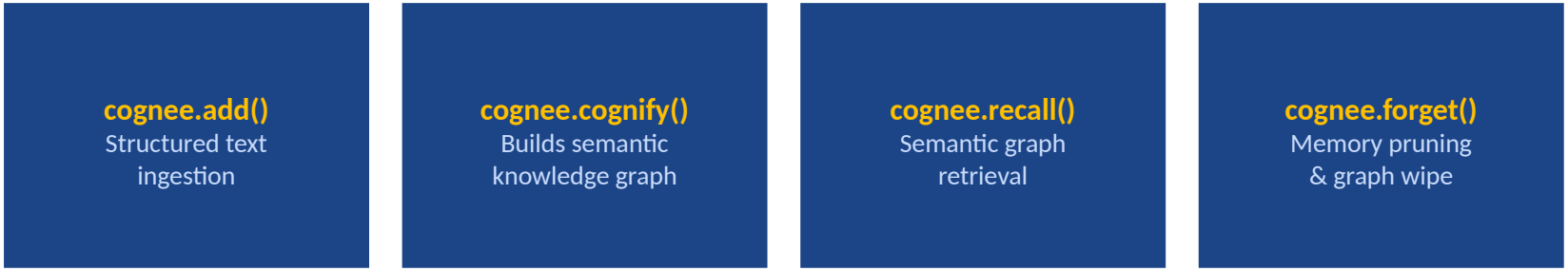
- Custom hybrid query pipeline cuts semantic retrieval latency down to ~1.5s.
- Bypasses standard slow LLM graph traversals for instant response times.
- Maintains dense context lookup (top_k=150) under peak load.

SYSTEM ARCHITECTURE & KEY ACTIVITIES

5-layer hybrid stack: React → Express → FastAPI → Cognee → SQLite/LanceDB



Core Cognee API Calls:



4. CUSTOMER RELATIONSHIPS

How Déjà establishes & maintains relationships with each customer segment

Students & Job Seekers

Interactive Feedback Loop

•Expectation: Dynamic, low-friction growth & habit tracking.
•Establish: Real-time journal ingestion (The Slate) with immediate background processing.
•Maintain: Active reinforcement. Thumbs up/down feedback on Oracle answers directly updates graph node weights in the backend, improving future Q&A recall accuracy.

Knowledge Workers

Long-Term Cognitive Extension

•Expectation: Secure, highly reliable, multi-year external brain.
•Establish: Dense context semantic graph indexing (top_k=150 context window) mapping complex project links.
•Maintain: Continuous semantic consistency. Keeps cross-referenced timeline records and graph notes synchronized over years, avoiding memory rot.

Wellness Seekers

Private Safeguard Sanctuary

•Expectation: Absolute data privacy, trust, and user sovereignty.
•Establish: Offline-first local mode (local SQLite & LanceDB) requiring no server connection.
•Maintain: GDPR-aligned 'right to be forgotten' via target memory branch wipes using cognee.forget(). Allows complete data pruning.

5. REVENUE STREAMS

Free-first consumer model funded by B2B corporate wellness licensing

Free Core Tier (Free Forever)

₹0 / month ● Consumer Mainstream Tier

- Local-only graph storage (SQLite) & vector embedding (LanceDB).
- Zero server hosting overhead for the company (client-side execution).
- Unlimited Slate journal entries & manual Oracle Q&A queries.

Pro Tier (The Game Changer)

₹149 / month ($1.99/mo) ● Cloud Premium

- Seamless Multi-Device Sync & secure roaming via Cognee Cloud.
- Proactive continuous Blindspots analysis pushing real-time Paradox Alerts.
- Advanced Oracle reasoning using premium high-scale LLM synthesis.

B2B2C Corporate Wellness Model (Primary Monetization Engine)

- EAP Integrations: Employers buy sponsored bulk seats of Déjà Pro for employee mental health, habit tracking & stress analysis.
- API Licensing: White-label semantic recovery API integration for wellness app partners & journaling communities.

6. KEY RESOURCES

Infrastructure that makes Déjà possible

Technology Stack

- Python FastAPI — memory bridge service
- Node.js Express — API gateway & auth
- React + Vite — glassmorphism frontend
- Cognee SDK — semantic graph engine
- SQLite + LanceDB — local vector storage
- Supabase — chronological DB & Auth

AI & Cognee Integration

- Cognee Open Source (Python library)
- Cognee Cloud — hosted semantic graphs
- LiteLLM wrapper — multi-LLM support
- Gemini Flash Lite — low-latency inference
- gemini-embedding-001 — 768-dim vectors
- Auth0 Device Flow — cloud provisioning

Human & IP Resources

- Concurrency-safe async queue (SQLite lock fix)
- Custom hybrid retrieval shortcut (10× speed)
- Blindspots algorithm (graph vs. timeline diff)
- Supabase RLS schema (multi-tenant security)
- Dockerized deployment for Hugging Face / Render
- Comprehensive test suite (cognee, recall, speed)

7. KEY ACTIVITIES — COGNEE DEEP DIVE

How Déjà maximises Cognee's capabilities across every feature

Journal Ingest Pipeline	User submits text → Node.js gateway → FastAPI bridge → cognee.add() classifies entities → cognee.cognify() builds knowledge graph → vectors stored in LanceDB
Oracle Query Pipeline	User asks question → cognee.recall(top_k=150) traverses graph → raw nodes returned → local LiteLLM analyzer summarizes → response in ~1.5s (bypasses default Cognee LLM round-trip)
Blindspots Detection	Supabase timeline entries fetched → Cognee graph node density computed → cosine vector distance analysis → chronological gaps & contradictions surfaced as 'paradox alerts'
Memory Safeguard	Target topic identified → cognee.forget() severs semantic edges and prunes vector space → or cognee.prune for full graph wipe → confirmation returned to UI

Production concurrency: Serialized async queue prevents SQLite write-lock contention during peak ingestion — ready for enterprise load.

8. KEY PARTNERS

Ecosystem that powers Déjà

Cognee AI

- Primary AI memory partner
- Open Source Python SDK (cognee library)
- Cognee Cloud — hosted semantic graph service
- Auth0 device-code flow for user provisioning

Google / Gemini

- LLM: gemini/gemini-3.1-flash-lite
- Embedding: gemini-embedding-001 (768-dim)
- Low-latency inference for Oracle responses
- Via LiteLLM multi-provider wrapper

Supabase

- PostgreSQL chronological timeline store
- Row-Level Security (RLS) multi-tenancy
- Built-in Auth for session management
- Anon key access from React frontend

Deployment & Infra

- Hugging Face Spaces — primary hosting
- Render — alternate cloud deployment
- Docker — unified multi-process container
- LanceDB — local vector storage engine

9. COST STRUCTURE

Lean AI-native architecture with managed infrastructure



Cognee Cloud API

Usage-based per query/ingest beyond free tier



LLM & Embedding API

Gemini Flash Lite — among lowest cost LLMs available



Supabase (Database)

Free tier covers early users; Pro ~$25/mo at scale



Hosting (HF / Render)

Free tier on Hugging Face Spaces; Render ~$7/mo

Development & DevOps

Open source toolchain; minimal licensing costs

Cost Model

~$0

bootstrapped MVP cost

Variable cost scales with user growth only

WHY DÉJÀ WINS — BEST USE OF COGNEE

Demonstrating the full power of Cognee in a real-world production system

1. Dual Local + Cloud Architecture

First app to seamlessly switch between Cognee Open Source (SQLite/LanceDB) and Cognee Cloud — proving the SDK's full versatility.

2. Auth0 Device Flow Integration

Extended Cognee's ecosystem with an automated onboarding script (auth_cognee.py) using Cognee's device code flow — zero user friction.

3. Graph-Timeline Comparison Engine

Blindspots cross-references Supabase's relational timeline against Cognee's structural node density — a novel use of graph topology for personal insight.

4. Production Concurrency Solutions

Solved SQLite write-lock contention via serialized async queue — a real enterprise-grade fix that makes Cognee viable under concurrent load.

Thank You

Déjà — Cognee-Powered Memory Recovery Engine

🧠 **Semantic Memory**
Personal AI memory graph
powered by Cognee

🔮 **Natural Language**
Ask anything about
your past in plain English

🔒 **Privacy First**
Local + Cloud modes;
right to be forgotten

GitHub · Hugging Face Spaces · CMR ThinkFest 3.0 — Team Déjà

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